

Transportation and Assignment Problems

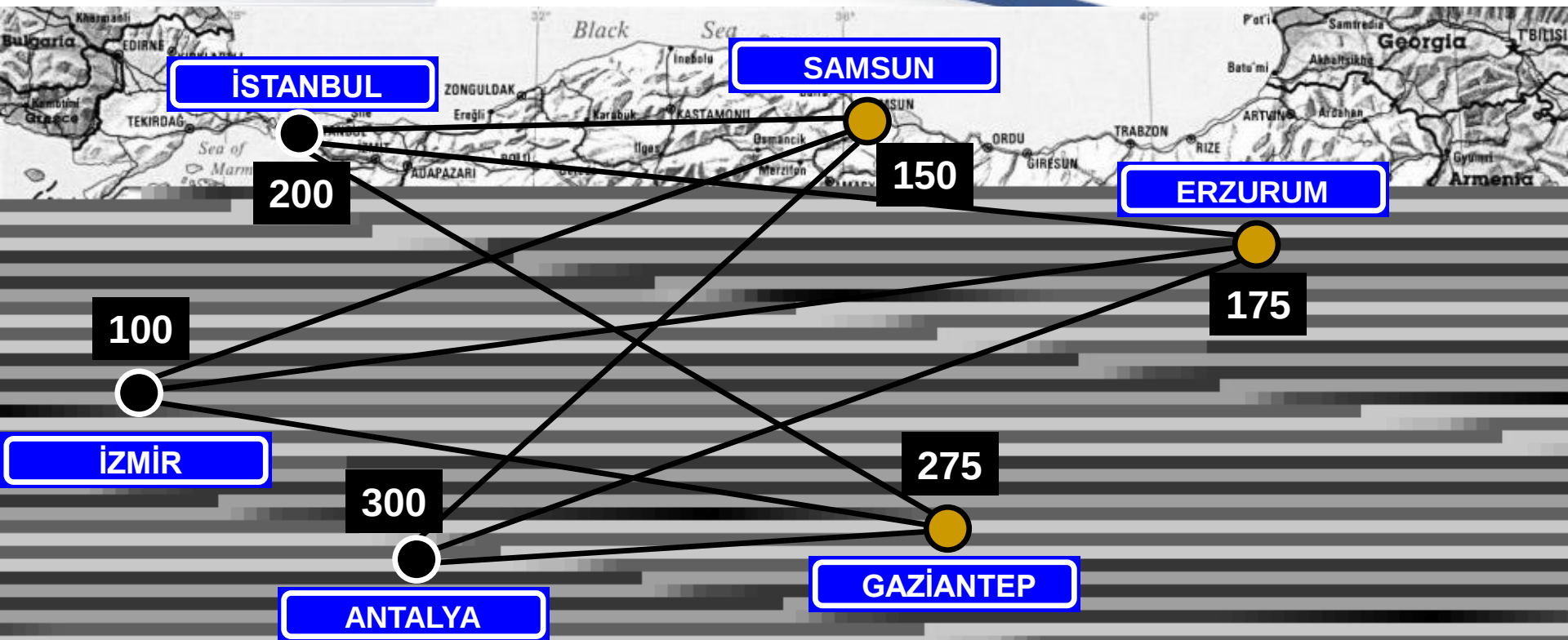


1. A product is transported from a number of sources to a number of destinations at the minimum possible cost.
2. Each source is able to supply a fixed number of units of the product, and each destination has a fixed demand for the product.



DEMAND (TONS)

SUPPLY (TONS)



Grain Elevator	Supply
1. Samsun	150
2. Erzurum	175
3. Gaziantep	275
Total	600 tons

Mill	Demand
A. İstanbul	200
B. İzmir	100
C. Antalya	300
Total	600 tons



Transportation Cost Relations (\$/ton)

Grain Elevator	Mill		
	A. İstanbul	B. İzmir	C. Antalya
1. Samsun	\$ 6	\$ 8	\$ 10
2. Erzurum	\$ 7	\$ 11	\$ 11
3. Gaziantep	\$ 4	\$ 5	\$ 12



Linear Programming Model

$$\text{minimize } Z = \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C}$$

subject to

$$x_{1A} + x_{1B} + x_{1C} = 150$$

$$x_{2A} + x_{2B} + x_{2C} = 175$$

$$x_{3A} + x_{3B} + x_{3C} = 275$$

$$x_{1A} + x_{2A} + x_{3A} = 200$$

$$x_{1B} + x_{2B} + x_{3B} = 100$$

$$x_{1C} + x_{2C} + x_{3C} = 300$$

$$x_{ij} \geq 0$$

From \ To				Supply
	A	B	C	
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600

In a **balanced transportation** model where supply equals demand, all constraints are equalities



The Transportation Tableau

From \ To				Supply
	A	B	C	
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600



Northwest Corner Method

Minimum Cell Cost Method

Vogel's Approximation Method - VAM



SS – Stepping-Stone Method

MODI – Modified Distribution Method



The Northwest Corner Method

To					Supply
From		A	B	C	
1		6	8	10	150
2		7	11	11	175
3		4	5	12	275
Demand		200	100	300	600



Initial Feasible Solution with Northwest Corner Method

$$\begin{aligned} Z &= \$6x_{1A} + 8x_{1B} + 10x_{1C} + 7x_{2A} + 11x_{2B} + 11x_{2C} + 4x_{3A} + 5x_{3B} + 12x_{3C} \\ &= 6(150) + 8(0) + 10(0) + 7(50) + 11(100) + 11(25) + 4(0) + 5(0) + 12(275) \\ &= \$5,925 \end{aligned}$$

The Initial Solution $Z = \$5,925$



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1. Allocate as much as possible to the cell in the upper left-hand corner, subject to the supply and demand constraints.
 2. Allocate as much as possible to the next adjacent feasible cell.
 3. Repeat step 2 until all rim requirements have been met.



The Minimum Cell Cost Method

From \ To				Supply
	A	B	C	
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600

The Initial Minimum Cell Cost Allocation



From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600

The Second Minimum Cell Cost Allocation



Initial Feasible Solution with Minimum Cell Cost Method

To					Supply
From		A	B	C	
1		6	8	10	150
			25	125	
2		7	11	11	175
				175	
3		4	5	12	275
		200	75		
Demand		200	100	300	600

The Initial Solution $Z = \$4,550$



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1. Allocate as much as possible to the feasible cell with the minimum transportation cost, and adjust the rim requirements.
 2. Repeat step 1 until all rim requirements have been met.



Transportation

Vogel's Approximation Method



Vogel's Approximation Method

From \ To	A	B	C	Supply	
1	6	8	10	150	2
2	7	11	11	175	4
3	4	5	12	275	1
Demand	200	100	300	600	
	2	3	1		

Penalty Cost is the difference between the lowest and the next lowest cell cost in a row (or column).

VAM allocates as much as possible to the minimum cost cell in the row or column with the largest penalty cost.

From \ To	A	B	C	Supply	
1	6	8	10	150	2
2	7	11	11	175	
3	4	5	12	275	1
Demand	200	100	300	600	
	2	3	2		



From \ To	A	B	C	Supply
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Demand	200	100	300	600



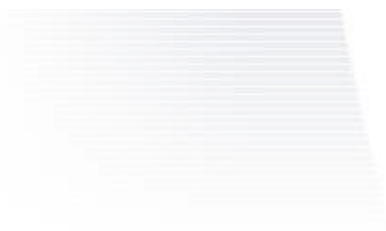


Initial Feasible Solution with Vogel's Approximation Method

To					
From		A	B	C	Supply
1		6	8	10	
				150	150
2		7	11	11	
		175			175
3		4	5	12	
		25	100	150	275
Demand		200	100	300	600

The Initial Solution $Z = \$5,125$



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1. Determine the penalty cost for each row and column by subtracting the lowest cell cost in the row or column from the next lowest cell cost in the same row or column.
 2. Select the row or column with the highest penalty cost (breaking ties arbitrarily or choosing the lowest-cost cell).
 3. Allocate as much as possible to the feasible cell with the lowest transportation cost in the row or column with the highest penalty cost.
 4. Repeat steps 1, 2, and 3 until all rim requirements have been met.
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The Unbalanced Transportation Model

An Unbalanced Model (Demand > Supply)

From \ To				Supply
	A	B	C	
1	6	8	10	150
2	7	11	11	175
3	4	5	12	275
Dummy	0	0	0	50
Demand	200	100	350	650



The Unbalanced Transportation Model

An Unbalanced Model (Demand < Supply)

From \ To	A	B	C	Dummy	Supply
1	6	8	10	0	150
2	7	11	11	0	175
3	4	5	12	0	375
Demand	200	100	300	100	700

